

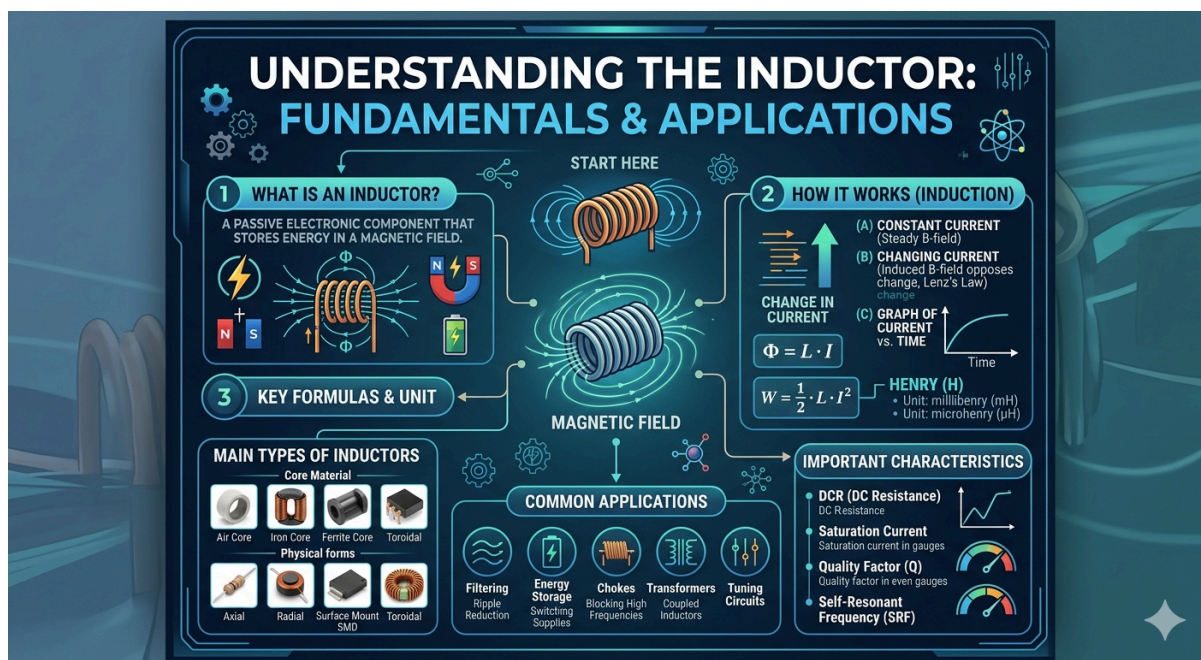
INDUCTOR FUNDAMENTALS: AN ENGINEERING DOCUMENTARY

Subject: Electromagnetic Theory and Application

**Classification: Technical Guide / Educational
Documentary Date: April 2026**

Sarkhan Babayev, Nazrin Sharbatli

**Verified by: Physics Teacher Azerbaijan Telman
Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)**



An inductor is a passive electronic component that stores energy in a **magnetic field** when electric current flows through it. While capacitors resist changes in voltage, inductors **resist changes in current**.

1. How It Works (The Physics)

When current passes through a coil of wire, it creates a magnetic flux (Φ). If the current changes, the magnetic field also changes, which induces an electromotive force (EMF) that opposes the initial change in current. This is known as **Lenz's Law**.

- **Steady Current:** The inductor acts like a simple wire (with minimal resistance).
 - **Changing Current:** The inductor generates a "back EMF" to fight the change.
-

2. Key Formulas & Units

The ability of an inductor to store energy is called **Inductance (\$L\$)**.

- **Unit of Measurement:** Henry (H), commonly found in millihenries (mH) or microhenries (μH).
- **Magnetic Flux:** $\Phi = L \cdot I$
- **Stored Energy:** $W = \frac{1}{2} \cdot L \cdot I^2$

3. Common Types of Inductors

Inductors are categorized by their core material and physical shape:

Type	Description	Common Use
Air Core	No magnetic core; wire is wound on a hollow form.	High-frequency applications (Radio/TV).
Iron Core	Higher inductance due to the magnetic properties of iron.	Audio equipment and power supplies.
Ferrite Core	Uses a ceramic-magnetic mix.	High-frequency suppression and EMI filtering.
Toroidal	Shaped like a donut to contain the magnetic field.	Transformers and high-efficiency power circuits.

4. Important Characteristics

When selecting an inductor, engineers look at these critical specs:

- **DCR (DC Resistance):** The internal resistance of the wire used in the coil.
- **Saturation Current:** The point where the core cannot hold any more magnetic flux.
- **Quality Factor (Q):** A measure of efficiency (energy stored vs. energy lost).
- **SRF (Self-Resonant Frequency):** The frequency at which the inductor starts acting like a capacitor.

5. Common Applications

- **Filtering:** Removing "ripple" from DC power supplies.
- **Energy Storage:** Used in switching regulators (like your phone charger).
- **Chokes:** Blocking high-frequency AC while letting low-frequency DC through.
- **Tuning:** Combined with capacitors to select specific radio frequencies.

Pro Tip: In a circuit diagram, an inductor is represented by a series of curly loops—mimicking the physical wire coil it actually is!